AI Shield - Lean Canvas

Unified AI fraud detection: scam messages, phishing URLs, emails

Karthik Siraparapu | Marwadi University | Internship ID IBMUEML004123 | AICTE-IBM SkillsBuild-BharatCares 2026

1. PROBLEM

- Millions of fraudulent SMS, WhatsApp messages, links and emails reach people every day (fake KYC, OTP, lottery, courier, UPI and 'digital arrest' scams).
- Rule-based spam filters miss newly worded scams; users cannot tell real from fake in seconds.
- Most detectors return a bare yes/no, so users do not learn why a message is dangerous.

EXISTING ALTERNATIVES

- Telecom / email spam filters
- Bank awareness SMS and helplines (1930, cybercrime.gov.in)
- Browser Safe Browsing blocklists (known URLs only)
- Asking a friend or searching the web

2. SOLUTION

- One web app with 3 ML modules: scam-text analyzer, phishing-URL checker, email checker.
- Explainable output: verdict, fraud probability, keyword and URL red flags.
- Honest validation: duplicate/leakage removal, hold-out tests, limits shown in the app.

3. UNIQUE VALUE PROPOSITION

- Paste any suspicious message, link or email and in seconds learn if it is a scam and exactly why.
- Free, open-source, built around Indian scam patterns (KYC, UPI, OTP, RBI/SBI impersonation).

HIGH-LEVEL CONCEPT

- 'Spell-check for scams': a second opinion before you click or pay.

4. UNFAIR ADVANTAGE

- Validation-first dataset audit (random labels and leakage found and fixed) that makes reported accuracy trustworthy.
- Open-source code plus an India-specific fraud lexicon that competitors must rebuild.
- Modular design: each model is swappable as better data arrives.

6. CUSTOMER SEGMENTS

- General smartphone users, students and first-time digital-payment users
- Senior citizens and their families
- Small businesses and shopkeepers using UPI
- Banks / fintechs and NGOs running fraud-awareness programmes (B2B)

EARLY ADOPTERS

- College students and cyber-safety clubs
- Family members who already forward 'is this real?' messages
- NGO / CSR fraud-awareness trainers

8. KEY METRICS

- Scam-text hold-out F1 (now 0.998) and recall on short messages (98.8% accuracy)
- URL ROC-AUC (now 0.81-0.83)
- Checks run per week; % flagged messages users confirm as scams (feedback button, planned)
- False-positive rate on genuine bank/OTP messages

5. CHANNELS

- Streamlit web app / link shared on WhatsApp and social media
- Colleges, cyber-awareness drives, NGOs (BharatCares network)
- GitHub and developer communities
- Future: browser extension, Android share-sheet, WhatsApp bot

7. COST STRUCTURE

- Cloud hosting (free tier now; low monthly cost at scale)
- Data acquisition and labelling of fresh Indian SMS/email scam samples (largest ongoing cost)
- Model retraining and monitoring; security review
- Developer time (currently student-built, unpaid)

9. REVENUE STREAMS (HYPOTHESES TO VALIDATE)

- Free public tool (impact first; supports SDG 9 and SDG 16)
- Paid API / white-label module for banks, fintech apps and telecom operators
- Institutional licences for colleges and corporate awareness training
- CSR / grant funding for awareness campaigns

Lean Canvas format by Ash Maurya (Creative Commons BY-SA 3.0). Metrics from this project's hold-out evaluation; business model items are hypotheses, not validated revenue.